Tutorial - 5

In this tutorial, you will install MPI and get acquainted with Message-Passing Interface (MPI) by running the hello-world program that is available on Moodle.

1. Install and test either OpenMPI or MPICH library on your computer.
2. Download the MPI hello-world program that is available on Moodle (from MPI Hello World folder) and compile and run the same. Convince yourself that the required number of messages are being printed depending on the number of processes you have requested.
3. Write an MPI program in which the host-process (designated as process with rank = 0) receives a single real (floating-point) value from all other processes and the host will add the received values and prints the final sum to the screen.
4. Refer to the trapezoidal rule program in which local values for a, b and n are used. Try parallelizing this version of the trapezoidal rule program using MPI. Use 2 or 4 processes to test.
5. Ponder how sieve of Eratosthenes can be implemented using MPI.